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📖 PUBLICATIONS (JOURNAL ARTICLES)

- 2025 **DentAll: Annotation-Efficient Dental Segmentation and Missing-Tooth Localization in Panoramic Radiographs**
Computers in Biology and Medicine (Under Review)
Authors: Shaily Bajpai, Hamza Shafiq, and Bumshik Lee
- 11/2025 **IA-MUNet: A Knowledge-Guided Instance-Aware Mamba-UNet for Efficient Multiclass Dental Image Segmentation**
Knowledge-Based Systems
Authors: Hamza Shafiq, Shaily Bajpai, and Bumshik Lee
DOI: <https://doi.org/10.1016/j.knosys.2025.114949>
- 10/2025 **DualBranchNetwork3D: A Hybrid CNN–Swin Transformer Model for Automated Cirrhotic Liver Segmentation on MRI**
IEEE Access
Authors: Siva Tanay Akash Jagarapu, Sashi Kiran Kaata, Shaily Bajpai, M. Raviraja Holla
DOI: <https://doi.org/10.1109/ACCESS.2025.3642267>

📖 PUBLICATIONS (CONFERENCE PAPERS)

- 2025 **Advanced Teeth Instance Segmentation Using Contextual Spatial Attention and Deeper Mask Head**
IEEE International Conference on Consumer Electronics (ICCE) / IEEE
Authors: Shaily Bajpai, Pankoo Kim, and Bumshik Lee
DOI: <https://doi.org/10.1109/ICCE63647.2025.10930043>
- 06/2024 **UNET-based Multiclass Teeth Segmentation in Panoramic Dental Radiographs**
Korea Communications Society Summer Conference (KICS)
Authors: Shaily Bajpai, and Bumshik Lee
- 06/2023 **Advances in Tooth Detection and Segmentation in Dental Panoramic X-ray Images: A Review**
Korea Communications Society Summer Conference (KICS)
Authors: Shaily Bajpai, and Bumshik Lee
- 11/2022 **Development of Ethernet/IP Adapter for Explicit Messaging in Cooperative Robot Communication**
22nd International Conference on Control, Automation and Systems (ICCAS)/ IEEE
Authors: Shaily Bajpai, Rahul Ranjan, Sungchul Lee, Kyung Oh Lee, Joongeup Kye
DOI: <https://doi.org/10.23919/ICCAS55662.2022.10003850>

📄 PATENTS

- 08/2025 **Deep Learning-Based Vision Inspection Device and Method (Application)**
Patent Issuer: Flexigo Co., Ltd. (Korean)
Shaily Bajpai and Lee Ki Yong
- 01/2025 **Device and Method for Segmenting Tooth Images in a Dental Diagnostic Device (Application)**
Patent Issuer: Chosun University Industry-Academia Cooperation Foundation (Korean)
Shaily Bajpai and Bumshik Lee

📄 POSTER PRESENTATION

- 01/2024 **Missing Teeth Count and Localization Methods in Dental Radiographs : A Review**
36th Workshop on Image Processing and Understanding (IPIU)
Shaily Bajpai and Bumshik Lee